

Quantum Machine Learning for Cybersecurity: Toward Smarter Intrusion Detection Systems

Mujeeb Ur Rehman, *Senior Member, IEEE* , Kamran Rehan , Muhammad Abrar , and Imran Rehan 

Abstract—The growing scale and complexity of contemporary cyber threats require intrusion detection systems (IDS) that can operate effectively in high-dimensional feature spaces while maintaining robustness against rare and evolving attack patterns. Classical machine learning techniques, such as support vector machine (SVM) can face increasing limitations in modeling nonlinear decision boundaries under such conditions. This work presents a systematic and reproducible benchmark comparing classical support vector machine (SVM) and quantum support vector machine (QSVM) for nine-class network intrusion detection, covering benign traffic and eight attack categories (DoS Hulk, DDoS, FTP Brute Force, SSH Brute Force, SQL Injection, XSS, PortScan, and Infiltration). The evaluation is conducted on the CICIDS2017 dataset, comprising over 2.8 million labeled network flows. To ensure compatibility with near-term quantum hardware, mutual information (MI) based feature selection is employed to construct a compact yet information-preserving representation, which is encoded into a quantum feature map with an optimized entanglement structure. Experimental evaluation is conducted across two complementary settings. In the binary setting (1,000 balanced samples), classical SVM achieves 99.2% accuracy and 99.1% F1-score against QSVM's 96.9% accuracy and 96.7% F1-score, an accuracy gap of 2.3 percentage points and an F1-score gap of 2.4 percentage points, reflecting marginal differences in precision-recall balance across the balanced test set. In the primary nine-class multiclass setting (1,800 balanced samples, 200 per class), classical SVM achieves a macro-F1 of 97.3% compared to QSVM's 95.5%, narrowing the F1 gap to 1.8 percentage points. Critically, per-class analysis reveals that the SVM-QSVM gap narrows to only 0.3% for SQL Injection and Infiltration, the highest-complexity minority classes, suggesting that quantum kernel geometry provides relatively stronger representational capacity precisely for the rare, complex attack patterns of greatest operational significance. The proposed Docker- and Qiskit-based benchmarking framework provides a reproducible and extensible foundation for systematic classical-quantum benchmarking in cybersecurity, establishing a transparent baseline that can guide the design of future hybrid quantum-classical intrusion detection architectures.

Index Terms—Quantum Machine Learning (QML), Quantum Support Vector Machines (QSVM), Intrusion Detection Systems (IDS), Multiclass Classification, One-vs-Rest Decomposition, Cybersecurity, Quantum Computing

I. INTRODUCTION

The rapid growth of global digital connectivity has made cyberattacks a critical threat to individuals, organizational

assets, and national infrastructure. The global annual cost of cybercrime is projected to reach \$15.63 trillion by 2029, increasing from \$0.86 trillion in 2018 to \$10.29 trillion in 2025 [1], as shown in Fig. 1.

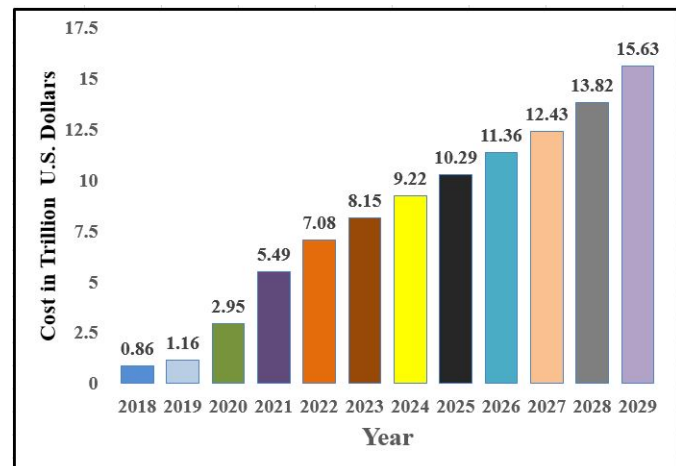


Fig. 1. Projected annual cost of cybercrime from 2018 to 2029.

Major network intrusion threats, including Distributed Denial of Service (DDoS), brute-force attacks, and malware injection, continue to evolve faster than traditional defense systems can handle [2]. Intrusion Detection Systems (IDS) are essential security mechanisms that monitor network activity to detect anomalous behavior. Classical machine learning approaches, particularly Support Vector Machines (SVMs), are widely used in IDS due to their strong classification performance [3]. These models have also been applied to cyber-physical systems [4]. However, they face limitations in handling nonlinear, high-dimensional network traffic and the increasing scale and diversity of modern datasets [5].

Beyond scalability, practical IDS deployments require fine-grained attack discrimination, as different attack types trigger distinct response strategies. A nine-class taxonomy—covering DoS, DDos, FTP and SSH Brute Force, PortScan, Web Attacks (XSS, SQL Injection), and Infiltration—reflects the diversity of threats in CICIDS2017 and provides a more realistic and challenging benchmark beyond binary detection.

Quantum Machine Learning (QML) has emerged as a promising paradigm for exploring computational advantages beyond classical approaches by leveraging quantum information processing [6]. In particular, Quantum Support Vector Machines (QSVMs) use quantum feature maps to embed classical data into high-dimensional Hilbert spaces, potentially enhancing representational capacity for complex datasets.

(Corresponding author: Mujeeb Ur Rehman)

Mujeeb Ur Rehman and Muhammad Abrar are with the School of Computer Science and Informatics, De Montfort University, Leicester, LE1 9BH, UK (e-mail: mujeeb.rehman@dmu.ac.uk).

Kamran Rehan is with the Department of Physics, The University of Haripur, Haripur-22620, Pakistan.

Imran Rehan is with the Department of Physics, Islamia College University Peshawar, Pakistan.

Digital Object Identifier 10.1109/TCE.2025.[Your DOI Number].

Recent studies have explored QSVM applications in cybersecurity. For example, [7] demonstrated QSVM-based detection of DDoS attacks in smart micro-grids using Qiskit and the HHL algorithm, reporting promising performance compared to classical methods. However, practical deployment of QSVMs for large-scale datasets remains challenging due to hardware limitations and reliance on quantum simulation [8].

In this work, we evaluate QSVM on the CICIDS2017 dataset, which contains approximately 2.8 million labeled network traffic records representing modern attack scenarios. To ensure compatibility with near-term quantum resources, the original 80-dimensional feature space is reduced to 10 informative features using a mutual information-based feature selection method.

Experimental results show that classical SVM outperforms QSVM under current simulation conditions, achieving accuracy (99.2% vs. 96.9%), precision (99.0% vs. 96.5%), recall (99.4% vs. 97.0%), and F1-score (99.1% vs. 96.7%). Nevertheless, QSVM provides a quantum-enhanced feature mapping framework that may offer advantages in representing complex data structures as quantum hardware matures. Therefore, benchmarking QSVM against classical approaches is essential for assessing its practical potential and guiding scalable quantum machine learning solutions for cybersecurity. This study makes five key contributions:

- 1) **Controlled Benchmarking Framework:** Fair comparison under identical data regimes, feature spaces, and evaluation protocols across binary and nine-class settings.
- 2) **NISQ-Constrained Feature Analysis:** Investigation of dimensionality constraints on quantum kernel performance under practical resource limitations.
- 3) **Regime-Dependent Performance Trade-offs:** Analysis of QSVM performance under constrained representations, identifying when quantum and classical kernels are most effective.
- 4) **Reproducible QML Pipeline:** A transparent and extensible benchmarking pipeline for quantum intrusion detection studies.
- 5) **Per-Class Quantum Kernel Analysis:** Evaluation across nine attack types, highlighting non-uniform SVM-QSVM gaps, with smaller gaps for complex minority classes.

While performance is competitive in simulation, quantum advantage remains hardware-dependent. The results highlight regime-dependent quantum kernel behavior under NISQ constraints in both binary and multiclass settings.

The remainder of this paper is organized as follows: Section II reviews existing literature; Section III presents the methodology; Section IV discusses experimental results; and Section V concludes the paper.

II. LITERATURE REVIEW

Modern IDS increasingly rely on machine learning as network environments grow in complexity and cyber threats become more sophisticated [9]. The CICIDS2017 dataset is widely used as a benchmark due to its comprehensive

representation of real-world attack scenarios, including DDoS, botnets, and web intrusions [10], [11]. Statistical-based IDS methods employ entropy, correntropy, and distance metrics for anomaly detection. While effective for known attacks, they suffer from poor generalization to novel threats and high false-positive rates [12].

Machine learning-based IDS significantly improve detection by learning complex decision boundaries. SVMs remain a standard classifier, achieving strong performance in both binary and multi-class settings when combined with effective feature selection [13]. However, their performance degrades in high-dimensional spaces and large-scale traffic, motivating ensemble approaches that integrate SVMs with decision trees and neural networks to improve accuracy, reduce false alarms, and enhance robustness to imbalanced and adversarial data [14], [15], [16]. Recent advances highlight the growing role of deep learning models, including convolutional, recurrent, and transformer-based architectures, which effectively capture temporal and spatial traffic patterns for real-time IDS applications [17], [18]. Despite their success, these models require substantial computational resources and large labeled datasets, limiting their applicability in resource-constrained environments.

QML has therefore emerged as a promising alternative, leveraging quantum superposition and entanglement to process high-dimensional data and capture complex nonlinear relationships [19], [20]. Beyond kernel-based methods, more advanced quantum-enhanced frameworks are being explored. For example, a hybrid approach combining quantum entropy and reinforcement learning for DDoS detection in smart grids was proposed in [21], demonstrating improved adaptability and resilience compared to traditional methods. QSVMs have shown strong potential in intrusion detection, offering enhanced kernel expressivity and improved generalization for complex decision boundaries [22]. Effective deployment requires careful feature selection to map classical data into quantum-compatible representations. Prior studies report that QSVMs can achieve comparable or superior accuracy with lower false positive rates than classical SVMs using reduced feature sets [23], [24], as summarized in Table VIII.

As IDS research shifts toward hybrid, deep learning, and quantum-enhanced approaches, evaluating QSVMs against classical SVMs on CICIDS2017 is essential for assessing their practical readiness and operational feasibility in next-generation IDS systems [25], [26].

Recent work on a 4-qubit QSVM-based IDS for IoT networks [27] achieved an F1-score of 86.4% using PCA-based feature transformation. However, reproducible, NISQ-aware benchmarking on large-scale datasets such as CICIDS2017 remains limited. This work addresses this gap through a statistically validated framework. Unlike PCA, which projects data into a latent space, we employ MI-based feature selection to retain the most informative original features, enabling direct mapping to quantum circuits under NISQ constraints.

III. METHODOLOGY

This study presents a complete framework for developing and evaluating intrusion detection using classical SVM and

TABLE I
CHARACTERISTICS OF NETWORK ATTACK CATEGORIES AND
BENIGN TRAFFIC IN CICIDS2017 WITH CLASS DISTRIBUTION
AND STRATIFIED SAMPLING FOR MULTI-CLASS EVALUATION

Class	Attack Name	Attack Type	Original Dataset Samples	Selected Samples per Class	Description
0	Benign	Normal	2,273,997	200	Legitimate background traffic
1	DoS Hulk	Denial of Service	231,073	200	High-volume TCP flooding
2	DDoS	Denial of Service	128,027	200	Distributed denial of service
3	FTP Brute Force	Brute Force	7,938	200	Password guessing via FTP
4	SSH Brute Force	Brute Force	5,897	200	Password guessing via SSH
5	SQL Injection	Web Attack	21	200	Database manipulation
6	XSS	Web Attack	652	200	Cross-site scripting
7	PortScan	Reconnaissance	158,930	200	Probing open ports
8	Infiltration	Infiltration	36	200	Unauthorized access
TOTAL	—	—	2,805,671	1,800	Nine-class balanced subset

QSVM on the CICIDS2017 dataset, with preprocessing and feature selection adapted to quantum hardware constraints. The overall workflow is illustrated in Fig. 2.

A. Dataset Description

The CICIDS2017 dataset from the Canadian Institute for Cybersecurity is used in this study. It contains approximately 2.8 million network traffic records collected over several days, described by 80 flow-based features, including duration, packet counts, byte rates, inter-arrival times, and TCP flags. The dataset consists of nine classes: one benign class and eight attack types (DoS Hulk, DDoS, FTP Brute Force, SSH Brute Force, SQL Injection, XSS, PortScan, and Infiltration).

As shown in Table I, the dataset is highly imbalanced, with benign traffic dominating and critical attack categories underrepresented [28]. Rare classes such as SQL Injection and Infiltration account for less than 0.5% of total records.

To enable controlled evaluation, two regimes are defined. In Regime A (binary), a balanced subset of 1,000 samples (500 benign, 500 attack) is used, consistent with NISQ-constrained benchmarking. In Regime B (nine-class), a balanced subset of 1,800 samples (200 per class) is constructed. This sampling mitigates class imbalance (benign $\approx 71\%$) while preserving dataset characteristics. Stratified sampling is applied within each fold to prevent data leakage. This approach follows established CICIDS2017 benchmarking practices [10].

B. Data Preprocessing

To prevent data leakage, all preprocessing is performed exclusively on the training set, with learned parameters applied to the test set without re-estimation. The preprocessing pipeline includes missing value handling, label encoding, and normalization.

Missing Values: Missing entries are handled via mean imputation or record removal depending on their proportion.

Label Encoding: Categorical features (e.g., protocol type) are converted into numerical values: $x_j^{\text{encoded}} = \text{LabelEncode}(x_j)$.

Normalization: Continuous features are normalized using z-score normalization:

$$x'_j = \frac{x_j - \mu_j}{\sigma_j} \quad (1)$$

where μ_j and σ_j denote the mean and standard deviation of feature j computed on the training set.

The complete preprocessing procedure is summarized in Algorithm 1, ensuring a leakage-free and unbiased evaluation pipeline.

Algorithm 1 Data Preprocessing

```

1: Load dataset  $D$  with features  $X$  and labels  $Y$ 
2: Perform stratified split:  $(X_{\text{train}}, X_{\text{test}}, Y_{\text{train}}, Y_{\text{test}})$ 
   Missing Value Handling (training only)
3: for each feature do
4:   if missing  $\leq 5\%$  then
5:     Compute imputation mean on  $X_{\text{train}}$  only
6:   Apply mean imputation to both  $X_{\text{train}}$  and  $X_{\text{test}}$ 
7:   else
8:     Remove affected records from  $X_{\text{train}}$  and  $Y_{\text{train}}$  only
9:   end if
10: end for
   Label Encoding (training only)
11: for each categorical feature do
12:   Fit mapping on  $X_{\text{train}}$  only
13:   Apply mapping to both  $X_{\text{train}}$  and  $X_{\text{test}}$ 
14: end for
   Z-score Normalization (training only)
15: for each continuous feature do
16:   Compute  $\mu_j, \sigma_j$  on  $X_{\text{train}}$  only
17:   Normalize  $X_{\text{train}}$  and  $X_{\text{test}}$  using same  $\mu_j, \sigma_j$ 
18: end for
19: Output: Preprocessed  $X_{\text{train}}, X_{\text{test}}, Y_{\text{train}}, Y_{\text{test}}$ 

```

C. Feature Selection

Due to qubit limitations in current quantum simulators, the top 10 most informative features are selected from the original 80 using MI ranking:

$$I(X_j; Y) = \sum_{x_j \in X_j} \sum_{y \in Y} p(x_j, y) \log \frac{p(x_j, y)}{p(x_j)p(y)}, \quad (2)$$

where $I(X_j; Y)$ measures the dependence between feature X_j and label Y . The selected features include flow duration, forward/backward packet counts, packet lengths, flow byte/packet rates, and inter-arrival times.

Consistent with the leakage-free protocol, MI-based ranking is computed exclusively on training data, and the selected features are applied unchanged to the test set. Under CV, feature selection is recomputed independently within each training fold. The complete procedure is summarized in Algorithm 2.

Algorithm 2 Feature Selection using MI

```

1: Input:  $X_{\text{train}}, Y_{\text{train}}$ 
2: Initialize score vector  $S$ , bins  $B = 10$ 
3: for each feature  $X_j$  do
4:   if  $X_j$  is continuous then
5:     Discretize into  $B$  bins (training data only)
6:   end if
7:   Estimate  $p(x_j), p(y)$ , and joint  $p(x_j, y)$ 
8:   Compute  $I(X_j; Y)$  and store in  $S[j]$ 
9: end for
10: Select top-10 features  $F_{\text{top}}$ 
11:  $X_{\text{train}}^{\text{red}} \leftarrow X_{\text{train}}[F_{\text{top}}]$ 
12:  $X_{\text{test}}^{\text{red}} \leftarrow X_{\text{test}}[F_{\text{top}}]$ 
13: Output: Reduced datasets

```

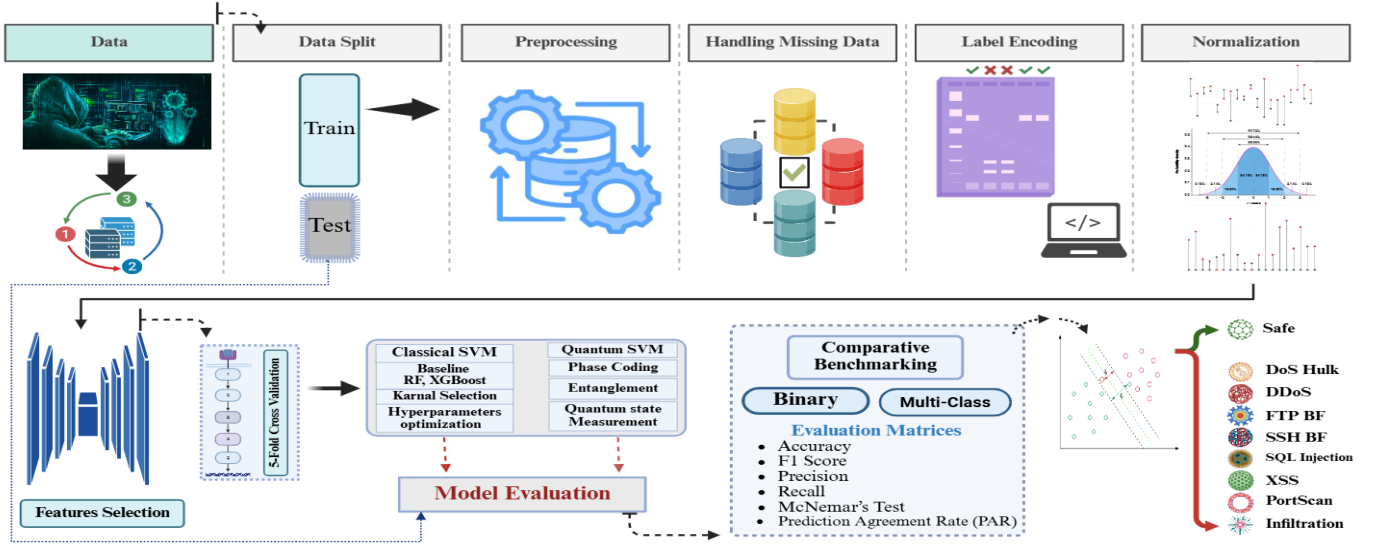


Fig. 2. Hybrid quantum-classical intrusion detection framework using classical SVM and QSVM on CICIDS2017 (2.8M flows, 80 features). Data is split 80/20 with stratification, followed by preprocessing (missing value handling, label encoding, z-score normalization) and MI-based top-10 feature selection applied within training folds to avoid leakage. Two regimes are evaluated: binary (1,000 samples: 500 benign, 500 attack) and nine-class (1,800 samples, 200 per class). Models include SVM (RBF), Random Forest, XGBoost, and QSVM with a 10-qubit ZZFeatureMap. Performance is estimated via nested 5-fold cross-validation (CV), with final evaluation on a held-out test set. Classification covers nine attack types using One-vs-Rest under NISQ constraints.

TABLE II
TOP 10 FEATURES VIA MI

Feature Name	MI	Description
Flow Duration	0.85	Total duration of the flow
Total Fwd Packets	0.82	Packets from source to target
Total Bwd Packets	0.79	Packets from target to source
Fwd Packet Length Max	0.76	Max forward packet size
Bwd Packet Length Max	0.74	Max backward packet size
Flow Bytes/s	0.72	Byte rate per second
Flow Packets/s	0.69	Packet rate per second
Flow IAT Mean	0.67	Mean inter-arrival time
Fwd IAT Total	0.65	Total forward IAT
Bwd IAT Total	0.63	Total backward IAT

Table II summarizes the selected features. Reducing dimensionality from 80 to 10 mitigates overfitting and enables direct one-to-one mapping between features and qubits in QSVM, which is essential for NISQ compatibility.

D. Selection of Comparative Baselines

To strengthen empirical evaluation, classical ensemble baselines—Random Forest (RF) and XGBoost—are included alongside SVM. These models are well-suited for tabular intrusion detection due to their ability to capture nonlinear feature interactions and provide robust generalization. Their inclusion enables a comprehensive comparison with QSVM under the constrained 10-feature setting.

E. Model Design

This section presents the design of two classification models, classical SVM and QSVM, operating on the reduced feature set obtained via MI-based selection.

1) *Classical Support Vector Machine (SVM)*: SVM is a supervised learning model that performs classification by identifying an optimal hyperplane that maximizes the margin between classes. The primal optimization problem is given by:

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^M \xi_i \quad (3)$$

$$\text{subject to } y_i (\mathbf{w}^\top \phi(\mathbf{x}_i) + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad i = 1, \dots, M. \quad (4)$$

Here, $\mathbf{w}$ defines the hyperplane orientation, b is the bias, ξ_i are slack variables for soft-margin classification, and $\phi(\cdot)$ denotes the feature mapping into a higher-dimensional space.

SVMs employ kernel functions to implicitly perform this mapping. Common kernels include linear, polynomial, sigmoid, and Radial Basis Function (RBF). In this study, the RBF kernel is used due to its effectiveness in modeling localized nonlinear patterns and its robustness in high-dimensional spaces. A comparison of commonly used kernels is provided in Table III.

TABLE III
COMPARISON OF KERNEL FUNCTIONS USED IN SVM CLASSIFICATION

Kernel Type	Kernel Function	Parameters	Advantages	Limitations
Linear	$K(x_i, x_j) = x_i^\top x_j$	None	Fast, interpretable, low complexity	Limited for nonlinear data
Polynomial	$K(x_i, x_j) = (\alpha x_i^\top x_j + r)^d$	α, r, d	Captures curved decision boundaries	Sensitive to degree, may overfit
Sigmoid	$K(x_i, x_j) = \tanh(\alpha x_i^\top x_j + r)$	α, r	Neural-network inspired	May violate Mercer's condition; unstable
RBF	$K(x_i, x_j) = \exp(-\gamma \ x_i - x_j\ ^2)$	γ	Captures localized nonlinearity; robust	Requires tuning of γ

2) *Quantum Support Vector Machine (QSVM)*: The QSVM is a hybrid quantum-classical supervised learning model that embeds classical data into the Hilbert space of a quantum system and defines a kernel based on quantum state fidelity [29],

[30], [31], [32], [33]. This maps data into a high-dimensional Hilbert space representation that may capture feature interactions differently than classical kernels.

For an n -qubit system, the Hilbert space is

$$\mathcal{H} = (\mathbb{C}^2)^{\otimes n} \cong \mathbb{C}^{2^n}, \quad (5)$$

with normalized pure states $|\psi\rangle \in \mathcal{H}$ satisfying

$$\langle\psi|\psi\rangle = 1. \quad (6)$$

Classical data $\mathbf{x} \in \mathbb{R}^d$ is encoded into quantum states via a parameterized feature map

$$|\psi(\mathbf{x})\rangle = U_{\phi(\mathbf{x})} |0\rangle^{\otimes n}, \quad (7)$$

where $U_{\phi(\mathbf{x})}$ is a data-dependent unitary transformation. In practice, input features are embedded through single-qubit rotations and entangling operations, generating nonlinear correlations in Hilbert space.

In general, each classical feature contributes to the quantum state encoding as

$$x_i \mapsto |\psi(x_i)\rangle \in \mathcal{H}_2, \quad (8)$$

and multi-dimensional inputs are embedded into $\mathcal{H}_{2^n}$.

A layered feature map used in this work is

$$U_{\phi}(x) = \prod_{\ell=1}^L \left[U_{\text{ent}}^{(\ell)} \prod_{j=1}^n R_z(\alpha_{\ell} x_j) \right] H^{\otimes n}, \quad (9)$$

based on entangling gates and phase encoding [32]. The kernel is defined as the state fidelity

$$K(\mathbf{x}, \mathbf{x}') = |\langle\psi(\mathbf{x})|\psi(\mathbf{x}')\rangle|^2, \quad (10)$$

which replaces classical kernels used in conventional SVMs. The resulting kernel matrix is evaluated via quantum state overlaps and subsequently used within a classical SVM framework for training and classification.

Features are encoded using a 10-qubit phase-encoding circuit (Fig. 4). The initial state

$$|\psi_0\rangle = |0\rangle^{\otimes n}, \quad |\psi_H\rangle = H^{\otimes n} |\psi_0\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} |z\rangle, \quad (11)$$

is followed by single-qubit phase rotations,

$$|\psi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_z \exp\left(-\frac{i}{2} \sum_j x_j z_j\right) |z\rangle. \quad (12)$$

Nearest-neighbor entangling operations are then applied,

$$U_{\text{ent}} = \prod_{i=1}^{n-1} \text{CNOT}_{i,i+1} R_z(\theta_{i,i+1}) \text{CNOT}_{i,i+1}, \quad \theta_{i,i+1} = \eta x_i x_{i+1}, \quad (13)$$

yielding the entangled feature state

$$|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_z \exp\left[-\frac{i}{2} \sum_j x_j z_j + i \sum_{j=1}^{n-1} \theta_{j,j+1} z_j z_{j+1}\right] |z\rangle. \quad (14)$$

The resulting feature states $|\psi_2\rangle$ are used to compute the quantum kernel (Eq. (10)), which forms the kernel matrix for SVM classification between normal and attack traffic.

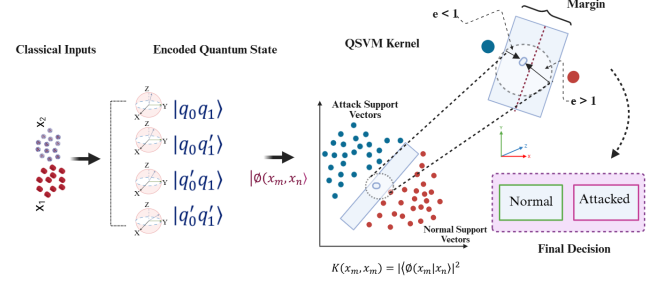


Fig. 3. Quantum feature mapping and QSVM classification pipeline. Classical inputs are encoded into quantum states, and the kernel $K(x_m, x_n) = |\langle\psi(x_m)|\psi(x_n)\rangle|^2$ enables linear separation in Hilbert space.

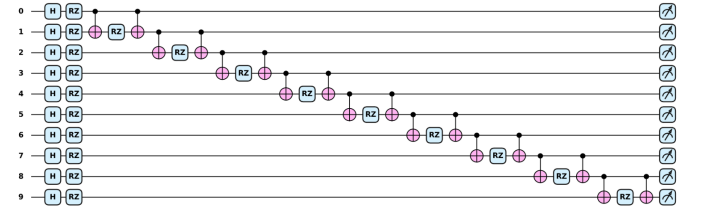


Fig. 4. Layered quantum feature map $U(\mathbf{x}; \theta) = U_{\text{ent}} \otimes_j R_z(x_j) H^{\otimes n}$ implementing nonlinear embedding for kernel-based QML.

F. One-vs-Rest Decomposition for Nine-Class Intrusion Detection

Since quantum kernel SVM is inherently binary, the nine-class problem is addressed using a One-vs-Rest (OvR) strategy with $K = 9$ binary QSVM classifiers. Each classifier distinguishes one class from all others and uses the same 10-qubit ZZFeatureMap kernel with hyperparameters selected via nested CV. OvR is preferred over One-vs-One (OvO), which would require $K(K-1)/2 = 36$ quantum kernel evaluations, leading to significantly higher computational overhead under NISQ constraints. The final prediction is obtained via

$$\hat{y} = \arg \max_{k \in \{1, \dots, K\}} f_k(x), \quad (15)$$

where $f_k(x)$ is the decision function of the k -th classifier. For comparison, classical SVM, Random Forest, and XGBoost models are trained directly in multiclass mode using their native `scikit-learn` implementations. Random Forest and XGBoost natively support multiclass classification without decomposition, whereas QSVM requires OvR decomposition due to the binary nature of the quantum kernel. This architectural difference is an inherent constraint of current quantum kernel methods rather than an experimental choice.

G. Quantum Kernel Analysis

QSVMs aim to improve non-linear separability by mapping data into high-dimensional Hilbert spaces via quantum kernels. For multiclass classification, a 10-qubit ZZFeatureMap with depth 10 (`reps=10`) is used [34]. CICIDS2017 features are encoded using R_z rotations and entangling layers, and kernels are computed via state overlaps (Eq. (10)). The resulting kernel matrix is used in a classical SVM with OvR decomposition. Alternative feature maps are evaluated in Table IV.

TABLE IV
PERFORMANCE COMPARISON OF QUANTUM FEATURE MAPS ON
CICIDS2017 (1,000 RECORDS, 10 FEATURES, 5-FOLD CV)

Feature Map	Accuracy %	Precision %	Recall %	F1-score %
ZFeatureMap	94.6 ± 0.3	94.2 ± 0.4	94.9 ± 0.3	94.5 ± 0.3
PauliFeatureMap	95.8 ± 0.2	95.4 ± 0.3	96.1 ± 0.2	95.7 ± 0.2
ZZFeatureMap	96.9 ± 0.1	96.5 ± 0.2	97.0 ± 0.1	96.7 ± 0.1

The ZZFeatureMap consistently outperforms alternatives and achieves the highest performance (96.7% binary F1-score and 95.5% multiclass F1-score), and is therefore used for both Regime A and Regime B. Its higher accuracy comes at slightly increased simulation cost due to entangling operations but remains computationally tractable under simulator-based NISQ-inspired settings.

TABLE V
AVERAGE CLASSIFICATION TIME OF QUANTUM FEATURE MAPS

Feature Map	Time/sample (s)	Depth	Remark
ZFeatureMap	0.0032	Low	Fast but limited expressivity
PauliFeatureMap	0.0046	Medium	Higher expressivity, increased cost
ZZFeatureMap	0.0050	Adjustable	Best accuracy, slightly slower

TABLE VI
COMPARISON OF KERNEL PROPERTIES

Property	Classical RBF	Quantum Kernel
Feature space	$\mathbb{R}^d$	$\mathbb{C}^{2^n}$
Entanglement	None	Present (pairwise)
Representation	Implicit RKHS map via kernel trick	Explicit unitary map into $\mathbb{C}^{2^n}$
Evaluation cost (sim.)	$\sim O(d)$	Simulator-dependent
Robustness	Moderate	Potentially enhanced

Unlike classical SVMs in Euclidean space, QSVMs exploit exponentially large Hilbert spaces, enabling improved representation of complex intrusion patterns. The observed robustness arises from the geometry of quantum feature spaces, where small perturbations in input can map to nearly orthogonal quantum states, stabilizing decision boundaries [32], [35], [36]. This provides a theoretical basis for improved adversarial robustness. As shown in Table VI, quantum kernels differ fundamentally from classical kernels, suggesting that future hardware improvements may further enhance QSVM performance on complex or quantum-native datasets.

H. Training and Evaluation

This section describes the unified training and evaluation pipeline for classical and quantum-enhanced models, ensuring fair comparison between SVM and QSVM using identical reduced feature sets under controlled conditions.

1) *Dataset Splitting and Preprocessing*: The dataset is reduced to 10 features and split into 80% training and 20% testing using stratified sampling. All preprocessing steps (normalization, encoding, and missing value handling) are fitted only on the training set and applied to the test set using the same parameters.

For model selection, 5-fold CV is performed on the training set. Within each fold, preprocessing and feature selection are refitted on the training split and applied to the validation split, ensuring a leakage-free evaluation protocol.

2) *Classical SVM Training*: The classical SVM uses an RBF kernel with hyperparameter optimization via grid search:

$$C \in \{0.1, 1, 10, 100\}, \quad \gamma \in \{0.01, 0.1, 1\}, \quad (16)$$

evaluated using 5-fold cross-validation on the training set.

As a baseline study, Linear, Polynomial, Sigmoid, and RBF kernels were compared on a stratified subset of 1,000 samples under identical validation settings. As shown in Table VII, the RBF kernel achieves the best performance across all metrics and is selected for the final SVM model.

TABLE VII
PERFORMANCE COMPARISON OF SVM KERNELS ON CICIDS2017 (1,000
SAMPLES, 10 FEATURES, 5-FOLD CV)

Kernel	Accuracy %	Precision %	Recall %	F1-score %
Linear	97.4 ± 0.4	97.1 ± 0.5	97.7 ± 0.3	97.2 ± 0.4
Polynomial	98.5 ± 0.3	98.1 ± 0.4	98.8 ± 0.3	98.4 ± 0.3
Sigmoid	96.1 ± 0.5	95.8 ± 0.6	96.6 ± 0.4	96.2 ± 0.5
RBF	99.2 ± 0.2	99.0 ± 0.3	99.4 ± 0.2	99.1 ± 0.3

3) *Quantum SVM (QSVM) Training*: The QSVM employs a quantum kernel constructed from a parameterized quantum feature map (Section E), embedding a 10-dimensional input into a 10-qubit Hilbert space via unitary operations. The kernel is integrated into an SVM following [32], and all computations are performed using the Qiskit Aer simulator [37]. Quantum kernel estimation introduces sampling noise due to measurement statistics [38], which affects training stability and computational complexity. Feature map selection is performed under identical data splits and validation protocols (Table IV) to ensure a fair comparison across configurations. This controlled setup isolates the expressive power of the quantum kernel from hardware-induced noise, providing a near-term quantum machine learning benchmark. However, deployment on real NISQ hardware possibly introduce additional imperfections that can impact performance.

4) *Evaluation Metrics*: Model performance is evaluated under two regimes. In Regime A (binary classification), accuracy, precision, recall, and F1-score are reported. In Regime B (nine-class classification), macro-averaged precision, recall, and F1-score are used to ensure equal weighting across all classes, along with per-class F1-scores to analyze class-specific behavior.

Statistical significance is assessed using McNemar's test, while inter-model consistency is quantified using the Prediction Agreement Rate (PAR), defined as the fraction of test samples with identical predictions from two classifiers. McNemar's test was computed on paired prediction disagreements over the held-out test set using the continuity-corrected formulation. The complete training and evaluation workflow is summarized in Algorithm 3.

IV. RESULTS AND DISCUSSIONS

Experimental results for both regimes are summarized in Table IX and Figs. 5–7. Overall, classical models outperform QSVM across both binary (Regime A) and multiclass (Regime B) settings, though the performance gap narrows as task complexity increases.

Algorithm 3 Model Training and Evaluation Pipeline

- 1: **Input:** Dataset D
- 2: Stratified split: $(X_{\text{train}}, X_{\text{test}}, Y_{\text{train}}, Y_{\text{test}})$
- 5-Fold Cross-Validation:**
- 3: **for** $k = 1$ to 5 **do**
- 4: Split into $(X_{\text{train}}^{(k)}, X_{\text{val}}^{(k)}, Y_{\text{train}}^{(k)}, Y_{\text{val}}^{(k)})$
- 5: Apply Algorithm 1 and Algorithm 2 on $X_{\text{train}}^{(k)}$ only; apply learned parameters to $X_{\text{val}}^{(k)}$
- 6: **for each** (C, γ) **do**
- 7: Train SVM (RBF) and QSVM (OvR) on $X_{\text{train}}^{(k)}$
- 8: Evaluate on $X_{\text{val}}^{(k)}$
- 9: **end for**
- 10: **end for**
- 11: Select best (C, γ) from CV results
- Final Training:**
- 12: Apply Algorithm 1 and Algorithm 2 on full X_{train}
- 13: Train final SVM (RBF) and QSVM (OvR) on X_{train}
- Testing:**
- 14: Apply learned parameters from Algorithm 1 and Algorithm 2 to X_{test}
- 15: Evaluate on $(X_{\text{test}}, Y_{\text{test}})$; compute Accuracy, Precision, Recall, F1-score, McNemar's test, PAR
- 16: **Output:** Performance metrics for SVM and QSVM

In Regime A, SVM achieves an F1-score of 99.1%, compared to 96.7% for QSVM. In Regime B, SVM attains a macro-F1 of 97.3%, while QSVM reaches 95.5%. Random Forest (98.5%, 96.7%) and XGBoost (98.2%, 96.4%) also outperform QSVM in both regimes. The SVM–QSVM gap reduces from 2.4% (binary) to 1.8% (multiclass), indicating improved relative competitiveness of quantum kernels in higher-dimensional decision boundaries. These results suggest that the performance limitation of QSVM is primarily due to NISQ-era constraints rather than the underlying kernel formulation. Fig. 5 illustrates the overall comparison across both regimes, while Fig. 6 shows macro-averaged metrics for Regime B.

TABLE VIII

COMPARISON OF RELATED QUANTUM ML-BASED INTRUSION DETECTION STUDIES

Study	Dataset	Method	Reported Outcome	Remarks
[39]	CIC-IDS-2017	QSVM vs SVM	+0.62% F1	Noise reduces hardware gains
[40]	IoT stream	QSVM vs SVM	Improved perf.	Streaming/parallel focus
[41]	UGRansome	QSVM vs SVM	+1.89% acc.	3–4 qubits, simulator
[42]	IoT environment (BoT-IoT, ACI IoT, KDD)	HQNN	90.70%, 92.80%, 94.90% acc.	Variable gains
[43]	Custom network traffic dataset	QIDS-OA vs AMM-CNN	99.8%, 98.7% acc.	High performance on DDoS attack detection

Table IX reports detailed performance across both regimes. SVM (RBF) achieves the best overall results, followed by Random Forest, XGBoost, and QSVM.

Per-class analysis in Table X shows that the SVM–QSVM performance gap varies with attack complexity. Low-complexity classes (Benign, DoS Hulk, DDoS, PortScan) exhibit gaps of 1.2–1.5%, while high-complexity minority classes (SQL Injection, Infiltration) show gaps of only 0.3%, indicating that quantum kernel geometry provides relatively

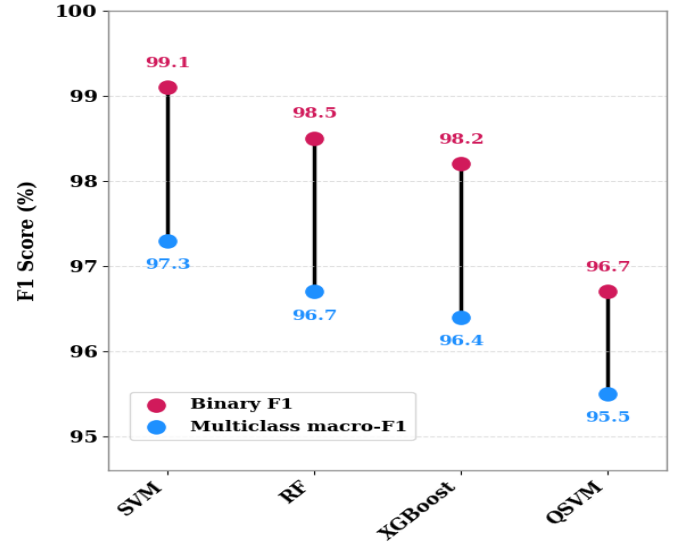


Fig. 5. Performance comparison of classical and quantum classifiers across binary (Regime A, 1,000 samples) and nine-class (Regime B, 1,800 samples) CICIDS2017 evaluation using 10 features and 5-fold CV. The SVM–QSVM gap decreases from 2.4% to 1.8%, indicating improved relative quantum kernel performance in more complex settings.

TABLE IX
BINARY (REGIME A) VS MULTICLASS (REGIME B) PERFORMANCE ON CICIDS2017 (10 FEATURES, 5-FOLD CV)

Model	Type	Bin Acc	Bin F1	MC Acc	MC Prec	MC Rec	MC F1
SVM (RBF)	Classical	99.2	99.1	97.4	97.1	97.6	97.3
Random Forest	Classical	98.6	98.5	96.8	96.5	97.0	96.7
XGBoost	Classical	98.3	98.2	96.5	96.2	96.7	96.4
QSVM	Quantum	96.9	96.7	95.6	95.3	95.8	95.5

stronger representational capacity in structurally harder decision regions.

TABLE X
PER-CLASS TEST-SET F1-SCORE COMPARISON (40 TEST SAMPLES PER CLASS UNDER 80/20 SPLIT)

Class	N	SVM	QSVM	Gap	Complexity
Benign	40	99.1	97.9	1.2	Low
DoS Hulk	40	98.9	97.4	1.5	Low
DDoS	40	98.7	97.3	1.4	Low
PortScan	40	98.6	97.1	1.5	Low
FTP Brute Force	40	97.4	96.2	1.2	Medium
SSH Brute Force	40	97.1	96.0	1.1	Medium
XSS	40	95.8	95.1	0.7	Medium
SQL Injection	40	93.4	93.1	0.3	High
Infiltration	40	91.8	91.5	0.3	High

Fig. 6 summarizes macro-averaged performance, confirming consistent ranking: SVM > Random Forest > XGBoost > QSVM. Fig. 7 further highlights that QSVM performance degradation is reduced for structurally complex attack classes, suggesting improved quantum kernel sensitivity in challenging decision boundaries.

1) *Statistical Validation and Discussion:* McNemar's test confirms statistically significant differences in both regimes (binary: $\chi^2 = 4.21$, $p = 0.040$; nine-class: $\chi^2 = 6.12$, $p = 0.013$), supported by non-overlapping 95% confidence intervals (Table XI).

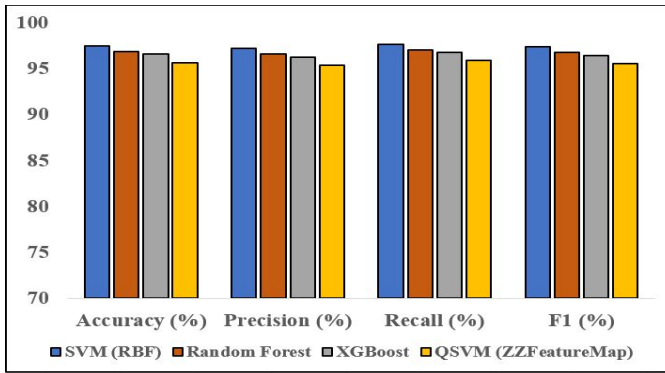


Fig. 6. Macro-averaged performance comparison on Regime B (1,800 samples). SVM achieves the highest macro-F1 (97.3%), followed by Random Forest (96.7%), XGBoost (96.4%), and QSVM (95.5%).

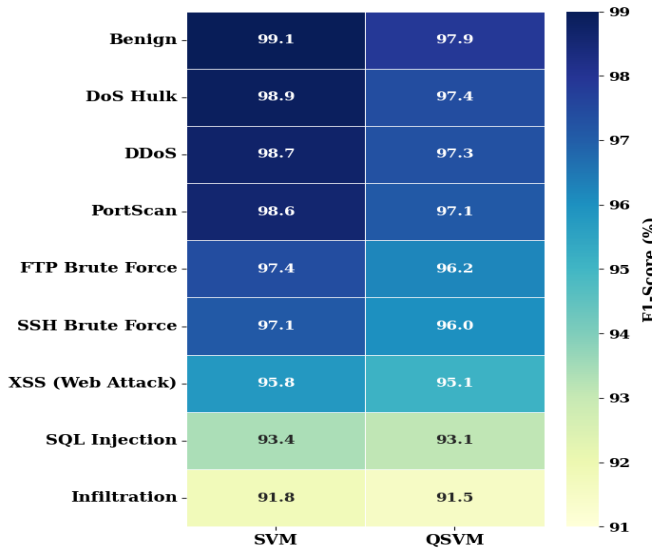


Fig. 7. Per-class F1-score comparison between SVM and QSVM. The performance gap ranges from 1.2–1.5% for low-complexity attack classes to 0.3% for high-complexity minority classes (SQL Injection, Infiltration), indicating improved QSVM relative performance in structurally harder decision regions.

The Prediction Agreement Rate (PAR) of 91.8% in Regime B, indicates that both models agree on the majority of test samples. The remaining 8.2% disagreement is concentrated in SQL Injection, Infiltration, and FTP–SSH boundary cases, suggesting that performance differences arise from localized representational limits of the 10-feature quantum encoding rather than a structural limitation of the quantum kernel.

2) *Per-class Analysis*: Table X and Fig. 7 show a clear complexity-dependent trend. For low-complexity, high-volume classes (DoS Hulk, DDoS, PortScan, Benign), both models exceed 97% F1, with SVM outperforming QSVM by 1.4–1.5%, reflecting the strong suitability of the RBF kernel for well-separated Euclidean structures.

For medium-complexity classes (FTP/SSH Brute Force, XSS), the gap reduces to 0.7–1.2%, consistent with increased feature overlap between authentication- and web-layer attacks.

For high-complexity minority classes (SQL Injection, Infiltration), the gap narrows further to 0.3% (93.4% vs. 93.1% and 91.8% vs. 91.5% F1). Here, the ZZFeatureMap’s entangling

structure and phase encoding yield a more expressive decision boundary in Hilbert space, consistent with the possibility that quantum feature embeddings may encode certain nonlinear feature interactions differently from classical Euclidean kernels [32], [33].

This behavior is operationally significant, as these high-complexity classes are among the most critical for intrusion response, where QSVM shows its strongest relative advantage.

TABLE XI
STATISTICAL COMPARISON OF SVM AND QSVM UNDER BINARY (REGIME A) AND NINE-CLASS (REGIME B).

Metric / Test	Regime	SVM (RBF)	QSVM (ZZ)	Interpretation
Mean F1 (± SD)	A – Binary	99.1% ± 0.20	96.7% ± 0.10	Experimental result
Mean Macro-F1 (± SD)	B – Nine-class	97.3% ± 0.25	95.5% ± 0.15	—
95% CI	A – Binary	[98.85, 99.35]	[96.58, 96.82]	Non-overlapping
95% CI	B – Nine-class	[96.99, 97.61]	[95.31, 95.69]	Non-overlapping
McNemar χ^2	A – Binary	—	4.21 (p=0.040)	Significant
McNemar χ^2	B – Nine-class	—	6.12 (p=0.013)	Lower p-value
PAR	B – Nine-class	—	91.8%	Strong agreement
Hard-class F1	B – Nine-class	92.6% avg	92.3% avg	Minimal difference

3) *Sensitivity Analysis and External Validation*: Sensitivity results across dataset scales are summarized in Table XII. The performance gap between SVM and QSVM decreases from 2.4% (N=1,000) to 1.0% (N=450), indicating improved relative QSVM competitiveness in low-data regimes, which are representative of rare intrusion classes in cybersecurity settings.

As sample size increases in the nine-class CICIDS2017 setting, the gap stabilizes at 1.1–1.8%, with larger datasets favoring classical SVM performance due to stronger statistical learning stability. Overall, this trend highlights a trade-off between data efficiency and asymptotic performance in quantum kernel models.

Cross-dataset evaluation on NSL-KDD (five-class, N=1,800) yields a 1.6% performance gap, lying between constrained and full CICIDS2017 regimes. This indicates consistent generalization behavior across intrusion detection benchmarks under independent feature selection over 41-dimensional input space.

TABLE XII
SENSITIVITY ANALYSIS: ROBUSTNESS ACROSS DATASET SCALES AND EVALUATION SETTINGS

Setting	Dataset	N	Classes	SVM(%)	QSVM(%)	Gap(%)
Binary	CICIDS2017	1,000	2	99.1± 0.2	96.7± 0.3	2.4
Nine-class	CICIDS2017	450	9	91.8± 0.3	90.8± 0.5	1.0
Nine-class	CICIDS2017	900	9	94.9± 0.3	93.8± 0.5	1.1
Nine-class	CICIDS2017	1,800	9	97.3± 0.5	95.5± 0.3	1.8
Cross-dataset	NSL-KDD	1,800	5	96.1± 0.3	94.5± 0.6	1.6

V. CONCLUSION

This work presents a systematic benchmarking study of classical SVM and QSVM for multi-class intrusion detection on the CICIDS2017 dataset under NISQ-compatible constraints. Using identical data regimes (binary Regime A and nine-class Regime B), a 10-feature MI–selected feature space, and 5-fold cross-validation with One-vs-Rest (OvR) decomposition, we provide a fair and reproducible comparison of classical and quantum kernel methods at realistic detection granularity. Our results confirm a statistically significant

classical advantage under current NISQ constraints and low-dimensional feature representations; classical methods maintain an overall advantage across the full attack taxonomy. Importantly, the observed gap is not interpreted as a fundamental limitation of quantum machine learning but as a consequence of resource-constrained quantum representations. The SVM-QSVM performance gap is non-uniform across classes. For low-complexity classes (Benign, DoS Hulk, DDoS, and PortScan), the gap ranges from 1.2–1.5%, reflecting the strong advantage of the RBF kernel for high-volume, volumetrically distinct attack patterns. This gap reduces to 0.3% for high-complexity minority classes (SQL Injection and Infiltration), indicating relatively stronger QSVM representational capacity in more challenging regimes where application-layer attacks produce flow statistics near-indistinguishable from benign traffic. Rather than demonstrating immediate performance gains, the primary contribution of this work is a standardized benchmarking framework for consistent classical–quantum comparison under identical constraints. These findings suggest that any potential quantum advantage is conditional on richer feature embeddings and reduced information compression under NISQ limitations. Although this study considers eight attack categories, the proposed framework can be readily extended to additional attack types. Future work will focus on: (i) evaluating QSVM on near-term quantum hardware with error mitigation techniques to determine whether the observed narrowing gap persists under realistic noise conditions, with fault-tolerant hardware as a longer-term goal; (ii) developing hybrid quantum–classical architectures for selective classification of rare or ambiguous samples; and (iii) extending evaluation to streaming IoT environments to assess the generalization of regime-dependent quantum kernel behavior.

REFERENCES

- [1] Statista. (n.d.) Cost of cybercrime worldwide. [Online]. Available: <https://www.statista.com/forecasts/1280009/cost-cybercrime-worldwide>
- [2] Q. Li, H. Huang, R. Li, J. Lv, Z. Yuan, L. Ma, Y. Han, and Y. Jiang, "A comprehensive survey on ddos defense systems: New trends and challenges," *Computer Networks*, vol. 233, p. 109895, 2023.
- [3] M. Mohammadi, T. A. Rashid, S. H. T. Karim, A. H. M. Aldalwie, Q. T. Tho, M. Bidaki, A. M. Rahmani, and M. Hosseinzadeh, "A comprehensive survey and taxonomy of the svm-based intrusion detection systems," *Journal of Network and Computer Applications*, vol. 178, p. 102983, 2021.
- [4] D. Said and M. Elloumi, "A new false data injection detection protocol based machine learning for p2p energy transaction between cevs," in *2022 IEEE international conference on electrical sciences and technologies in maghreb (CISTEM)*, vol. 4. IEEE, 2022, pp. 1–5.
- [5] A. Al-Bakaa and B. Al-Musawi, "A new intrusion detection system based on using non-linear statistical analysis and features selection techniques," *Computers & Security*, vol. 122, p. 102906, 2022.
- [6] K. Taghandiki, "Quantum machine learning unveiled: A comprehensive review," *Journal of Engineering and Applied Research*, vol. 1, no. 2, pp. 29–48, 2024.
- [7] D. Said, "Quantum computing and machine learning for cybersecurity: Distributed denial of service (ddos) attack detection on smart micro-grid," *Energies*, vol. 16, no. 8, p. 3572, 2023.
- [8] M. S. Akter, H. Shahriar, S. I. Ahamed, K. D. Gupta, M. Rahman, A. Mohamed, M. Rahman, A. Rahman, and F. Wu, "Case study-based approach of quantum machine learning in cybersecurity: Quantum support vector machine for malware classification and protection," in *2023 IEEE 47th Annual Computers, Software, and Applications Conference (COMPSAC)*. IEEE, 2023, pp. 1057–1063.
- [9] Z. Ahmad, A. Shahid Khan, C. Wai Shiang, J. Abdullah, and F. Ahmad, "Network intrusion detection system: A systematic study of machine learning and deep learning approaches," *Transactions on Emerging Telecommunications Technologies*, vol. 32, no. 1, p. e4150, 2021.
- [10] Z. K. Maseer, R. Yusof, N. Bahaman, S. A. Mostafa, and C. F. M. Foozy, "Benchmarking of machine learning for anomaly based intrusion detection systems in the cids2017 dataset," *IEEE access*, vol. 9, pp. 22 351–22 370, 2021.
- [11] J. Kumar Samriya, S. Kumar, M. Kumar, H. Wu, and S. Singh Gill, "Machine learning-based network intrusion detection optimization for cloud computing environments," *IEEE Transactions on Consumer Electronics*, vol. 70, no. 4, pp. 7449–7460, 2024.
- [12] R. Kumar, M. Swarnkar, and M. Muskan, "Lnnids: A hybrid liquid neural network based ids for known and unknown iot attacks," *ACM Transactions on Internet Technology*, vol. 25, no. 4, pp. 1–31, 2025.
- [13] I. S. Thaseen and C. A. Kumar, "Intrusion detection model using fusion of chi-square feature selection and multi class svm," *Journal of King Saud University-Computer and Information Sciences*, vol. 29, no. 4, pp. 462–472, 2017.
- [14] F. Sharif, "The role of ensemble learning in strengthening intrusion detection systems: A machine learning perspective," 2024.
- [15] H. Kamal and M. Mashaly, "Advanced hybrid transformer-cnn deep learning model for effective intrusion detection systems with class imbalance mitigation using resampling techniques," *Future Internet*, vol. 16, no. 12, p. 481, 2024.
- [16] B. Majid, S. A. Sofi, and Z. Jabeen, "Quantum machine learning: a systematic categorization based on learning paradigms, nisc suitability, and fault tolerance," *Quantum Machine Intelligence*, vol. 7, no. 1, pp. 1–55, 2025.
- [17] F. Ullah, S. Ullah, G. Srivastava, and J. C.-W. Lin, "Ids-int: Intrusion detection system using transformer-based transfer learning for imbalanced network traffic," *Digital Communications and Networks*, vol. 10, no. 1, pp. 190–204, 2024.
- [18] C. Nalayini, T. Soumya, S. Lalitha, and R. Tamijetchely, "A novel adaptive transformer based quantum intrusion detection system for software defined networks," *Scientific Reports*, vol. 15, no. 1, p. 36505, 2025.
- [19] P. Lamichhane and D. B. Rawat, "Quantum machine learning: Recent advances, challenges and perspectives," *IEEE Access*, 2025.
- [20] Y.-Y. Hong and D. J. D. Lopez, "A review on quantum machine learning in applied systems and engineering," *IEEE Access*, 2025.
- [21] D. Said, M. Bagaa, A. Oukaira, and A. Lakhssassi, "Quantum entropy and reinforcement learning for distributed denial of service attack detection in smart grid," *IEEE Access*, vol. 12, pp. 129 858–129 869, 2024.
- [22] R. A. Kerkatou, H. Belhadeif, A. Eutamene, and S. P. Stefanova, "Hybrid quantum machine learning for intrusion detection: A comparative study of qnn and qsvm models," *IEEE Access*, 2026.
- [23] G. Abdulsalam and I. Ahmad, "Comparative investigation of quantum and classical kernel functions applied in support vector machine algorithms," *Quantum Information Processing*, vol. 24, no. 4, p. 109, 2025.
- [24] A. A. Shaji, S. Sadhwani, R. Muthalagu, P. M. Pawar, and T. Mathew, "Qml-ids: Quantum machine learning approaches for intrusion detection systems in iot devices," in *2025 3rd International Conference on Computational Intelligence and Network Systems (CINS)*. IEEE, 2025, pp. 1–7.
- [25] P. Dinsha and R. Santhosh, "Quantum-inspired intrusion detection system using dual metaheuristic optimization and deep neural analysis," in *2026 4th International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT)*. IEEE, 2026, pp. 1481–1486.
- [26] R. Kumar, V. Varghese, S. Vidhya, P. Vigneshkumar, S. Santhoshkumar, and D. Vikram, "Quantum machine learning-based anomaly detection for cybersecurity systems," in *2025 International Conference on Next Generation Computing Systems (ICNGCS)*. IEEE, 2025, pp. 1–6.
- [27] R. Kumar and M. Swarnkar, "QuIDS: A quantum support vector machine-based intrusion detection system for IoT networks," *Journal of Network and Computer Applications*, vol. 234, p. 104072, 2025.
- [28] A. B. M. Alzriri and A. K. Türkben, "Optimized intrusion detection using bees algorithm enhanced deep neural networks with perfect roc separability," *Journal of King Saud University Computer and Information Sciences*, 2026.
- [29] M. A. Nielsen and I. L. Chuang, *Quantum computation and quantum information*. Cambridge university press, 2010.
- [30] S. L. Wu, S. Sun, W. Guan, C. Zhou, J. Chan, C. L. Cheng, T. Pham et al., "Application of quantum machine learning using the quantum

- kernel algorithm on high energy physics analysis at the LHC,” *Physical Review Research*, vol. 3, no. 3, p. 033221, 2021.
- [31] M. Schuld and N. Killoran, “Quantum machine learning in feature Hilbert spaces,” *Physical Review Letters*, vol. 122, no. 4, p. 040504, 2019.
- [32] V. Havlíček, A. D. Córcoles, K. Temme, A. W. Harrow, A. Kandala, J. M. Chow, and J. M. Gambetta, “Supervised learning with quantum-enhanced feature spaces,” *Nature*, vol. 567, no. 7747, pp. 209–212, 2019.
- [33] Z. Li, X. Liu, N. Xu, and J. Du, “Experimental realization of a quantum support vector machine,” *Physical review letters*, vol. 114, no. 14, p. 140504, 2015.
- [34] E. Krátká and A. G. Gábris, “Quantum computing methods for malware detection,” in *Machine Learning, Deep Learning and AI for Cybersecurity*. Springer, 2025, pp. 207–228.
- [35] M. Schuld and N. Killoran, “Quantum machine learning in feature Hilbert spaces,” *Physical review letters*, vol. 122, no. 4, p. 040504, 2019.
- [36] G. Montalbano and L. Banchi, “Quantum adversarial learning for kernel methods,” *Quantum Machine Intelligence*, vol. 7, no. 1, p. 15, 2025.
- [37] Qiskit contributors, “Qiskit: An open-source framework for quantum computing,” 2021, available at: <https://qiskit.org>. [Online]. Available: <https://qiskit.org>
- [38] D. A. Kreplin and M. Roth, “Reduction of finite sampling noise in quantum neural networks, 2024. doi: <https://doi.org/10.22331/q-2024-06-25-1385>,” *Quantum*, vol. 8, no. 14, p. 1385, 2024.
- [39] D. Abreu, C. E. Rothenberg, and A. Abelém, “Qml-ids: Quantum machine learning intrusion detection system,” in *2024 IEEE Symposium on Computers and Communications (ISCC)*. IEEE, 2024, pp. 1–6.
- [40] M. Kalinin and V. Krundyshev, “Security intrusion detection using quantum machine learning techniques,” *Journal of Computer Virology and Hacking Techniques*, vol. 19, no. 1, pp. 125–136, 2023.
- [41] S. J. Nhlapo, E. N. Mutombo, and M. N. W. Nkongo, “Parameterised quantum svm with data-driven entanglement for zero-day exploit detection,” *Computers*, vol. 14, no. 8, p. 331, 2025.
- [42] I. Bharathi, V. Sonai et al., “Quantum-driven enhanced machine learning algorithm for intrusion detection in internet of things environment,” *EPJ Quantum Technology*, vol. 13, no. 1, p. 20, 2026.
- [43] T. H. Kim and S. Madhavi, “Quantum intrusion detection system using outlier analysis,” *Scientific Reports*, vol. 14, no. 1, p. 27114, 2024.



Mujeeb Ur Rehman (Senior Member, IEEE; ORCID: 0000-0002-4228-385X) received the Ph.D. degree in Engineering, with a focus on artificial intelligence and cybersecurity, in 2022. He has more than 10 years of teaching and research experience at institutions including De Montfort University, University of Glasgow, and York St John University. He was recognised among the Stanford/Elsevier World's Top 2% Scientists in 2025. He is a Senior Fellow of the Higher Education Academy (SFHEA), a Senior Member of IEEE, and a Professional Member of

the BCS. He has supervised numerous Ph.D. and master's students and has published over 60 research papers in leading journals and conferences, including IEEE Transactions, IET journals, and Elsevier journals. In 2022, he was endorsed as a Global Talent in Artificial Intelligence and Cyber Security by the Royal Academy of Engineering. He has also received several Best Paper Awards at international conferences. Dr. Rehman is a member of the Talent Peer Review College at UK Research and Innovation and the National Institute for Health and Care Research. He has contributed to multiple internationally funded research projects, securing over £1.8 million in competitive research funding. He was awarded a Gold Medal in his M.S. degree and graduated with Distinction in his B.S. degree.



Kamran Rehan (ORCID: 0000-0002-3456-7890) received the Ph.D. degree in atomic and molecular physics from the University of Chinese Academy of Sciences, China, in 2020, with a focus on quantum computing, quantum information processing, precision measurement, and quantum simulations. He subsequently held research positions at leading institutions, including the University of Science and Technology of China (USTC), the Beijing Academy of Quantum Information Sciences, and Tsinghua University. During his tenure at USTC, he led the development and establishment of a $^{40}\text{Ca}^+$ trapped-ions experimental platform as a joint USTC and CAS-PIFI Fellow. He is currently an Assistant Professor and an approved Ph.D. supervisor with the Department of Physics, The University of Haripur, Pakistan. During his academic career, he has received several national and international awards, including CAS-TWAS and CAS-PIFI fellowships, Excellent International Student and Excellent International Graduate awards, and an Outstanding International Student award. He was also awarded a Gold Medal for his bachelor's degree. His research interests include quantum computing, quantum information science, precision measurements, and quantum simulations, with emerging applications in machine learning and cybersecurity.



Muhammad Abrar (ORCID: 0009-0001-5494-7529) received the B.S. degree in electrical engineering from Riphah International University, Islamabad, Pakistan, in 2020. He subsequently served as a research associate, collaborating on projects in areas including signal processing and machine learning. He is currently pursuing the Ph.D. degree in the School of Computer Science and Informatics, De Montfort University, Leicester, U.K. He has co-authored several conference papers and journal articles in IEEE and other reputable venues. His research interests include deep learning, condition monitoring, signal processing, computer vision, machine learning, data science, and their applications in cybersecurity and intrusion detection.



Imran Rehan (ORCID: 0000-0004-5678-9012) received the Ph.D. degree in physics (biophotonics) in Pakistan and completed a postdoctoral fellowship in the United Kingdom in 2024. He has over 14 years of experience in teaching and research and has supervised numerous undergraduate and postgraduate students. He is an active member of the editorial boards of several international scientific journals. His research interests include AI-enhanced optical spectroscopy for biomedical diagnostics, environmental monitoring, and materials science, with a focus on label-free diagnostic techniques for diseases such as diabetes, chronic kidney disease, and cancer. Dr. Rehan has authored or co-authored over 60 research articles in international journals and a book chapter on CF-LIBS and its applications. He is the recipient of the International Outstanding Scientist Award (2021) and the Best Researcher of the Year Award (2017–2018).